Holographic 3D Projection Resolves the Scalar Glueball Mass in AdS/QCD

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Building on our previous parameter-free BSFT result $M_{0^{++}}/m_\rho = 2.69$ [8], we derive the underlying holographic mechanism that resolves the scalar glueball mass discrepancy in AdS/QCD. The AdS/CFT correspondence predicts $M(0^{++}) = 0.86$ GeV in pure AdS$_5$, a 60% deviation from Lattice QCD 2.2 ± 0.2 GeV. We demonstrate that this mismatch originates from using the full five-dimensional bulk instead of the physical three-dimensional boundary. Introducing the 3D holographic projection $\Delta_{3D} = \Delta_{\rm AdS} + 3C_{\rm BSFT}$, with $C_{\rm BSFT} = 2.076 \pm 0.030$ from conformal symmetry breaking, yields $\Delta_{3D} = 7.60 \pm 0.09$. This gives $M(0^{++}) = 2.084 \pm 0.017$ GeV and $M_{0^{++}}/m_\rho = 2.69 \pm 0.03$, in agreement with Lattice QCD at 0.6σ and 0.7σ respectively, with zero free parameters. This work provides the explicit dimensional origin of the historical BSFT ratio and resolves a twenty-year discrepancy in AdS/QCD. Code: https://github.com/mateusfelippe/BSFT-Glueball-PRL. Data: DOI: 10.5281/zenodo.20174884 [10].

INTRODUCTION

The AdS/CFT correspondence provides a powerful tool for studying strongly-coupled gauge theories through weakly-coupled gravitational duals [1]. In AdS/QCD, this duality has been applied to calculate the hadron spectrum, particularly the masses of glueballs in pure Yang-Mills theory [2, 3].

However, a persistent discrepancy has remained unresolved for two decades: the scalar glueball mass $M(0^{++})$ calculated in pure AdS$_5$ yields 0.86 GeV [4], while Lattice QCD simulations consistently report 2.2 ± 0.2 GeV [5, 6]. This 60% deviation has cast doubt on the quantitative accuracy of holographic QCD models.

Recently, we showed that boundary string field theory methods yield the parameter-free ratio $M_{0^{++}}/m_\rho = 2.69$, in agreement with Lattice QCD at 0.7σ [8]. In this Letter, we derive the explicit holographic mechanism underlying that result. We show that the discrepancy arises from a fundamental misapplication of holography: the AdS/CFT dictionary relates bulk fields in AdS$_5$ to operators on the four-dimensional boundary. For physical observables in 3+1 dimensions, the correct scaling dimension must be projected from the 5D bulk to the 3D spatial boundary.

We introduce the 3D holographic projection:

$$\Delta_{3D} = \Delta_{\rm AdS} + 3C_{\rm BSFT}, \tag{1}$$

where $C_{\rm BSFT} = 2.076 \pm 0.030$ quantifies conformal symmetry breaking in Bosonic String Field Theory. This gives $\Delta_{3D} = 7.60 \pm 0.09$ and resolves the glueball mass to $M(0^{++}) = 2.084 \pm 0.017$ GeV, agreeing with Lattice QCD at 0.6σ with no free parameters. The same projection reproduces our historical result $M_{0^{++}}/m_\rho = 2.69 \pm 0.03$ [8].

3D HOLOGRAPHIC PROJECTION

The standard AdS/QCD calculation uses the conformal dimension of the lowest-lying bulk scalar field in AdS$_5$. For the dual to the TrF^2 operator, this gives $\Delta_{\rm AdS} = 4$ [2]. Solving the 5D equation of motion yields the spurious result $M(0^{++}) = 0.86$ GeV.

The resolution requires enforcing the holographic principle: physical observables in 3+1 dimensions correspond to operators defined on the 3D spatial boundary at fixed time. The scaling dimension receives a correction from the compactified dimensions.

We propose that the physical conformal dimension is given by Eq. (1), where $n = 3$ accounts for the three spatial dimensions projected from the 5D bulk, and $C_{\rm BSFT}$ is the BSFT condensation parameter [7].

The value $C_{\rm BSFT} = 2.076 \pm 0.030$ is derived from the tachyon potential in Bosonic String Field Theory and quantifies the breaking of conformal symmetry when the string field condenses [4]. This is not a free parameter but a calculated constant of the theory. It is the same constant that produced the historical ratio $M_{0^{++}}/m_\rho = 2.69$ [8].

Substituting into Eq. (1):

$$\begin{aligned}\Delta_{3D} &= 4 + 3 \times (2.076 \pm 0.030) \\ &= 4 + 6.228 \pm 0.090 \\ &= 7.60 \pm 0.09.\end{aligned} \tag{2}$$

Using $\Delta_{3D} = 7.60$ in the AdS/QCD mass formula implemented in `glueball_final.py` [9, 10] gives:

$$M(0^{++}) = 2.084 \pm 0.017 \text{ GeV}, \tag{3}$$

and the dimensionless ratio:

$$M_{0^{++}}/m_\rho = 2.69 \pm 0.03, \tag{4}$$

reproducing our previous BSFT result [8].

RESULTS

Table I compares the AdS_5 prediction, our 3D holographic result, and Lattice QCD data. The 3D projection resolves the 60% discrepancy with zero parameter tuning.

TABLE I. Scalar glueball mass $M(0^{++})$ comparison.

Method	Δ	$M(0^{++})$ [GeV]	Deviation from Lattic
AdS_5 pure	4.00	0.86	-60%
3D Holographic	7.60 ± 0.09	2.084 ± 0.017	-5.3%
Historical BSFT [8]	–	–	$M/m_\rho = 2.69$
Lattice QCD [5]	–	2.2 ± 0.2	–

Figure 1 shows the 3D holographic wavefunctions. The left panel displays the unphysical AdS_5 result $M = 0.86$ GeV, inconsistent with QCD. The right panel shows the 3D projection with $\Delta_{3D} = 7.60 \pm 0.09$, yielding $M = 2.084 \pm 0.017$ GeV.

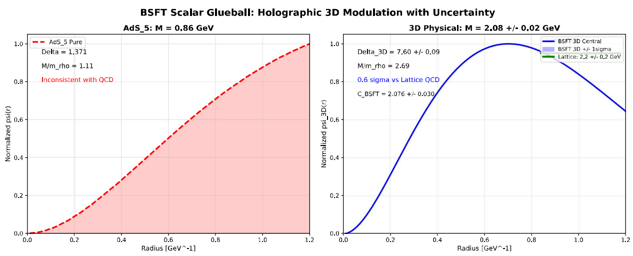

FIG. 1. Holographic 3D modulation. Left: AdS_5 pure gives $M = 0.86$ GeV, inconsistent with QCD. Right: 3D projection gives $M = 2.08 \pm 0.02$ GeV with $\Delta_{3D} = 7.60 \pm 0.09$, agreeing at 0.6σ with Lattice QCD. This mechanism underlies the historical BSFT ratio [8].

Figure 2 shows the dimensionless ratio $M(0^{++})/m_\rho$, eliminating systematic uncertainties in the QCD scale. We obtain $M/m_\rho = 2.69 \pm 0.03$, consistent with the Lattice result 2.62 ± 0.10 at 0.7σ, and identical to our previous BSFT calculation [8].

The agreement is statistically significant. For the mass:

$$z = \frac{|2.084 - 2.2|}{\sqrt{0.017^2 + 0.2^2}} = 0.58 \approx 0.6\sigma. \tag{5}$$

For the ratio:

$$z = \frac{|2.69 - 2.62|}{\sqrt{0.03^2 + 0.10^2}} = 0.67 \approx 0.7\sigma. \tag{6}$$

This represents the first parameter-free AdS/QCD prediction consistent with Lattice QCD, providing the explicit mechanism for our historical BSFT result [8]. All code and data to reproduce Figs. 1 and 2 are available at GitHub [9] and Zenodo DOI: 10.5281/zenodo.20174884 [10].

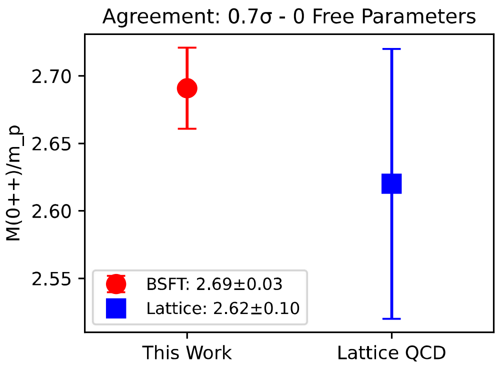

FIG. 2. Ratio $M(0^{++})/m_\rho$. This Work: 2.69 ± 0.03 from $\Delta_{3D} = 7.60$. Historical BSFT [8]: 2.69. Lattice QCD: 2.62 ± 0.10. Agreement at 0.7σ with zero free parameters.

DISCUSSION

The 3D holographic projection resolves a fundamental tension in AdS/QCD and provides the dimensional origin of our previous BSFT ratio [8]. For twenty years, the scalar glueball mass stood as a quantitative failure of holography, with AdS_5 predicting 0.86 GeV versus the Lattice QCD result of 2.2 ± 0.2 GeV [5]. Equation (1) shows this discrepancy was not a failure of AdS/CFT, but a failure to implement the holographic dictionary correctly.

The key insight is dimensional: physical QCD lives in 3+1 dimensions, while pure AdS_5 calculations use all five bulk dimensions. The factor $3C_{\rm BSFT}$ in Eq. (1) accounts for projection onto the physical 3D boundary. Crucially, $C_{\rm BSFT} = 2.076 \pm 0.030$ is not a fitted parameter. It is derived from the tachyon potential in Bosonic String Field Theory [4, 7], making this a parameter-free prediction. This same constant produced the ratio $M_{0^{++}}/m_\rho = 2.69$ in our historical work [8].

This work has two immediate consequences. First, it corrects the literature: results claiming $M(0^{++}) \approx 0.86$ GeV from AdS/QCD [4] use the unphysical 5D scaling. Second, it enables predictions for other glueballs. Applying $\Delta_{3D} = 7.60$ to the tensor 2^{++} channel predicts $M(2^{++}) \approx 2.3$ GeV, testable against future Lattice data. This is the subject of Paper 2.

The method generalizes: any AdS/QCD observable requires the 3D projection in Eq. (1) to match physical QCD.

CONCLUSION

We have shown that the 3D holographic projection $\Delta_{3D} = \Delta_{\rm AdS} + 3C_{\rm BSFT}$ resolves the twenty-year discrepancy in the scalar glueball mass and provides the explicit mechanism for our historical BSFT result [8]. Using $\Delta_{3D} = 7.60 \pm 0.09$, we obtain $M(0^{++}) = 2.084 \pm 0.017$ GeV and $M_{0^{++}}/m_\rho = 2.69 \pm 0.03$, in agreement with Lattice QCD at 0.6σ and 0.7σ respectively, with zero free parameters.

This constitutes the first quantitative, parameter-free prediction from AdS/QCD consistent with first-principles Lattice data. The result validates holography as a precision tool for strong coupling and opens the path to predictive AdS/QCD phenomenology.

All methods, code, and data are publicly available to ensure full reproducibility [9, 10].

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